

SNEVAM - Technical Requirements Document (TRD)

Project Type: 3D Fashion & Accessories E-commerce Platform

Objective: Build a scalable, SEO-friendly 3D commerce experience.

Technology Stack

Frontend: Next.js, React, TypeScript, Tailwind CSS, ShadCN UI, GSAP

3D: Three.js, React Three Fiber, Drei

Backend: NestJS, Node.js

Database: PostgreSQL, Redis

Infrastructure: Vercel, AWS S3, CloudFront

Functional Requirements

Authentication, Product Management, 3D Product Viewer, Cart, Checkout, Wishlist, Reviews, Admin Dashboard, Analytics.

API Requirements

Authentication APIs, Product APIs, Cart APIs, Order APIs, Wishlist APIs, and Admin APIs.

Database Schema

Users, Products, Orders, Cart, Reviews with PostgreSQL relational design.

Performance Requirements

LCP < 2.5s, FID < 100ms, CLS < 0.1. Target 60 FPS on desktop and 30+ FPS on mobile.

Security

JWT Authentication, HTTPS, Rate Limiting, CSRF Protection, Password Hashing, PCI-compliant payments.

Future Enhancements

AR Try-On, AI Personal Shopper, Loyalty Program, Mobile Apps, Multi-Vendor Marketplace.